

Failure / Authorization Test Evidence

Purpose

These three tests verify that the agent and application layer fail safely when given unauthorized tools, unknown case records, or incomplete tool arguments. They were run against the University Student Support Case Agent (Group G) using the command `python src/main.py`, with Groq as the primary model (`openai/gpt-oss-120b`). The agent is authorized to use two tools only: `get_case_status` and `create_support_ticket`.

Test 1: Unauthorized Tool Request

Purpose: Verify that the application rejects a tool that is not in the authorized tool catalogue.

Test input: Tool requested: `delete_student_record`, Case ID: `CASE-1045`

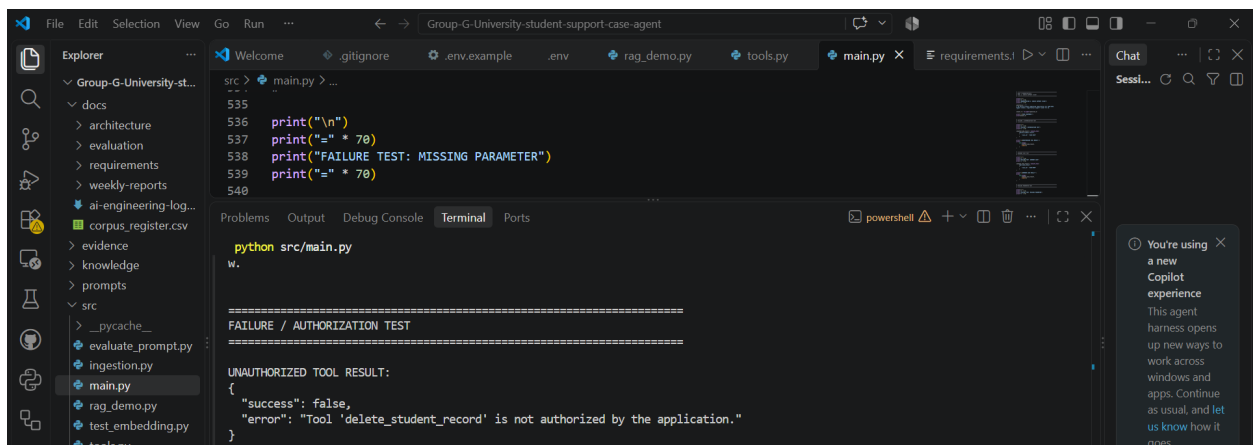
Observed result:

```
{
  "success": false,
  "error": "Tool 'delete_student_record' is not authorized by the application."
}
```

Expected behavior: The application must reject unauthorized tools and must not execute any operation outside the approved tool set.

Result: PASS

Evidence: Screenshot of the FAILURE / AUTHORIZATION TEST section showing the UNAUTHORIZED TOOL RESULT.



```
src > src/main.py > ...
535
536 print("\n")
537 print("=" * 70)
538 print("FAILURE TEST: MISSING PARAMETER")
539 print("=" * 70)
540

python src/main.py
w.

=====
FAILURE / AUTHORIZATION TEST
=====

UNAUTHORIZED TOOL RESULT:
{
  "success": false,
  "error": "Tool 'delete_student_record' is not authorized by the application."
}
```

Test 2: Unknown Case ID

Purpose: Verify that the `get_case_status` tool safely handles a case ID that does not exist in the authorized synthetic case data.

Test input: CASE-9999

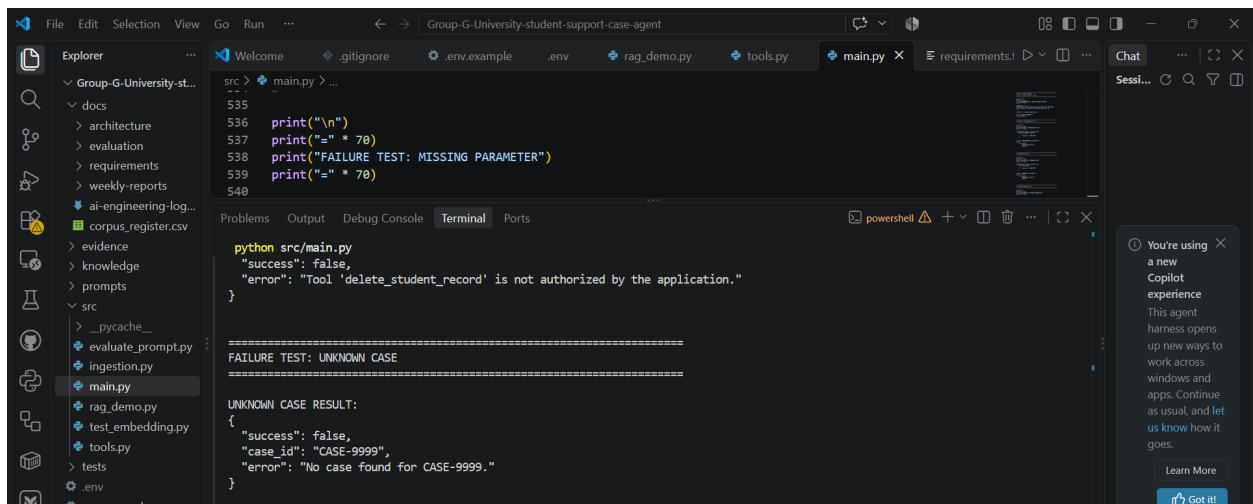
Observed result:

```
{
  "success": false,
  "case_id": "CASE-9999",
  "error": "No case found for CASE-9999."
}
```

Expected behavior: The application must not invent case information or return a successful result for an unknown case.

Result: PASS

Evidence: Screenshot of the FAILURE TEST: UNKNOWN CASE section showing the returned error.



```
python src/main.py
{"success": false,
 "error": "Tool 'delete_student_record' is not authorized by the application."
}
```

```
=====
FAILURE TEST: UNKNOWN CASE
=====

UNKNOWN CASE RESULT:
{
  "success": false,
  "case_id": "CASE-9999",
  "error": "No case found for CASE-9999."
}
```

Test 3: Missing Required Parameter

Purpose: Verify that the `create_support_ticket` tool rejects an incomplete request when the required `description` parameter is missing.

Test input:

```
{
  "case_id": "CASE-1045",
  "category": "registration"
}
```

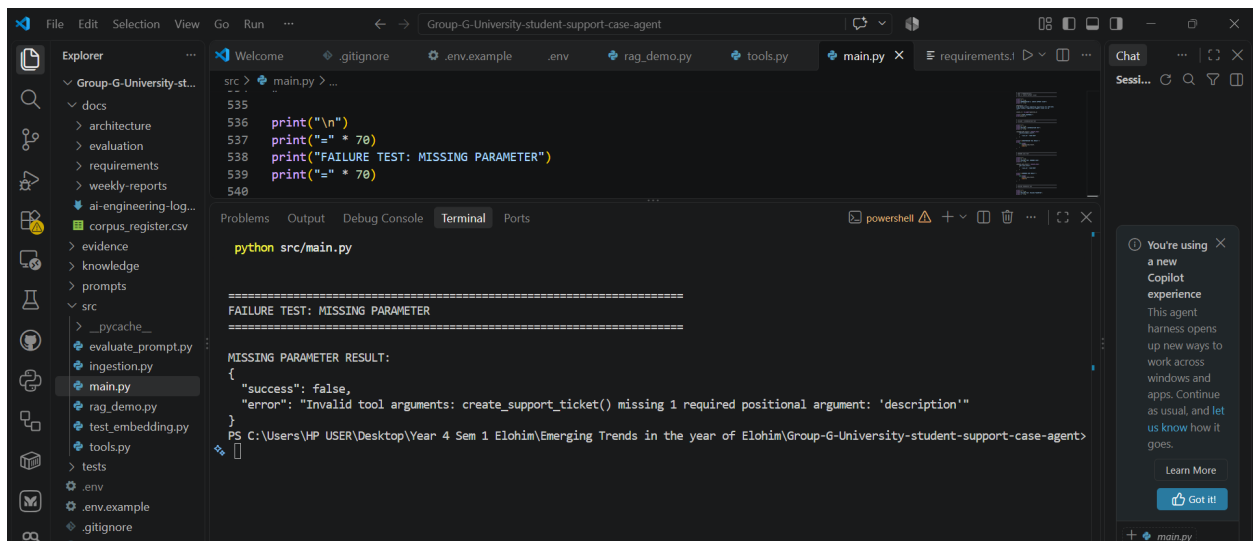
Observed result:

```
{
  "success": false,
  "error": "Invalid tool arguments: create_support_ticket() missing 1 required positional argument: 'description'"
}
```

Expected behavior: The application must not create a support ticket when a required parameter is missing.

Result: PASS

Evidence: Screenshot of the FAILURE TEST: MISSING PARAMETER section showing the returned error.



```
File Edit Selection View Go Run ... Group-G-University-student-support-case-agent
Explorer
  Group-G-University-st...
  docs
  > architecture
  > evaluation
  > requirements
  > weekly-reports
  > ai-engineering-log...
  corpus_register.csv
  > evidence
  > knowledge
  > prompts
  src
  > _pycache_
  evaluate_prompt.py
  ingestion.py
  main.py
  rag_demo.py
  test_embedding.py
  tools.py
  tests
  .env
  .env.example
  .gitignore
  README.md

src > main.py > ...
535
536 print("\n")
537 print("=" * 70)
538 print("FAILURE TEST: MISSING PARAMETER")
539 print("=" * 70)
540

Problems Output Debug Console Terminal Ports
python src/main.py

=====
FAILURE TEST: MISSING PARAMETER
=====

MISSING PARAMETER RESULT:
{
  "success": false,
  "error": "Invalid tool arguments: create_support_ticket() missing 1 required positional argument: 'description'"
}
PS C:\Users\HP USER\Desktop\Year 4 Sem 1 Elohim\Emerging Trends in the year of Elohim\Group-G-University-student-support-case-agent>
```

Summary

All three failure conditions were handled safely, and every test passed.

Test	Failure condition	Expected behavior	Observed result	Status
1	Unauthorized tool	Reject unauthorized operation	Tool rejected	PASS
2	Unknown case	Do not invent case data	Case not found	PASS
3	Missing parameter	Reject incomplete tool call	Missing parameter rejected	PASS

The application layer enforces the authorized tool set, returns an explicit error for records that do not exist, and validates required arguments before any action is taken.